Technology Report and Presentation

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Student: Luis Guilherme de Barros Andrade Faria - A00187785

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Lecturer: Dr. Dyvia Leekha

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Cloud Computing for ABC Enterprise: Key Contributions to Business Automation

1. Executive Summary

Cloud computing has become a core business utility - email, storage, and smart devices all run on someone else's infrastructure, delivered on demand (Nishimura, 2022). For ABC Enterprise from our case study, this model translated directly into results: start-up IT costs dropped by ~80% and the platform absorbed a 10x customer surge within a month, without a matching increase in headcount (Eliaçık, 2022; McHaney, 2021). This report outlines three key benefits of cloud adoption, the main challenges to manage, and a practical recommendation for service model, deployment model, cost model, and cloud provider. Where ABC needs a provider beyond "XYZ", AWS is recommended for its maturity, global reach, and automation-ready managed services (Amazon Web Services, n.d.).

2. Benefits of Cloud Computing vs. Traditional IT Infrastructure

Traditional IT means owning everything - servers, cooling, procurement cycles, and the staff to keep it running (McHaney, 2021). For a high-growth start-up like ABC, that model is a strategic handicap. Cloud flips it: instead of buying capacity, ABC rents capability, aligned to the NIST essential characteristics of on-demand self-service, measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011).

2.1. Cost Efficiency and Pay-as-you-grow Model

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no idle hardware costs (Eliaçık, 2022). ABC's ~80% reduction in start-up IT costs is the "measured service" characteristic in action: pay for compute-hours, storage GB-months, and data transfer actually consumed (Mell & Grance, 2011). As McHaney (2021) notes, operational overhead - personnel, training, upgrades, and security - often exceeds hardware cost over time, making pay-as-you-go a structural advantage. Beyond the bill, standard tasks like backups, patching, and scaling can be codified and automated, reducing human toil across the delivery pipeline (Accenture Technology, 2020).

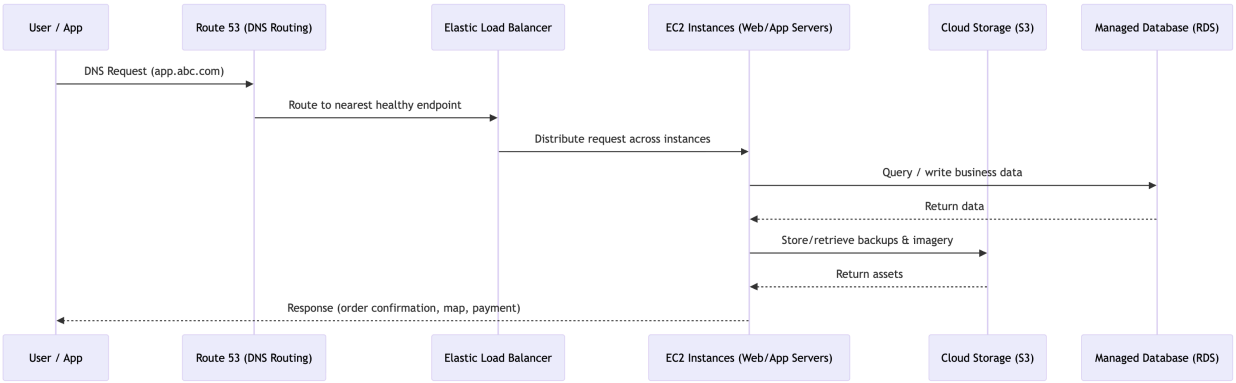

Figure 1 – ABC Enterprise cloud traffic flow — from user request through Route 53, load balancing, compute, storage, and database layers

2.2. Rapid Scalability for Business Growth

A 10x customer surge in a single month exposes the core weakness of on-premises infrastructure: procurement lead times mean hardware arrives after the opportunity has passed (McHaney, 2021). Cloud's rapid elasticity lets ABC scale compute up during

campaigns or city launches and scale down when demand normalizes - automatically, not reactively (Mell & Grance, 2011). Eliaçık (2022) highlights that this "endless scalability" protects against outages from traffic spikes, keeping latency low and availability high. For an app handling delivery, taxi bookings, and payments, that's not a `nice-to-have` - it's a **retention strategy**.

2.3. Reduced IT Management Overhead

In on-premises environments, more customers mean more infrastructure, which means more people to maintain it (McHaney, 2021). Cloud breaks that linear relationship. Through resource pooling and virtualization, providers consolidate physical resources across tenants - Manvi and Shyam (2021) describe this as partitioning, isolation, and consolidation that allows multiple workloads to share hardware without interference. ABC borrows that operational maturity: automated maintenance, managed monitoring, and resilient architectures that would be expensive to replicate in-house (Accenture Technology, 2020).

3. Challenges of Cloud Migration and Mitigation Strategies

Cloud adoption is not risk-free. Three challenges are most relevant for ABC (Eliaçık, 2022):

1. **Security and Privacy:** ABC handles payments and customer PII. Eliaçık (2022) notes that entrusting sensitive data to a third party means breaches may not be detected immediately. **Mitigation:** least privilege IAM, mandatory MFA, encryption at rest/in transit, and continuous alerting. The shared responsibility model means the provider secures the cloud infrastructure — ABC must secure what it deploys on top.

2. **Cost Volatility:** Pay-as-you-go can spiral without guardrails - overprovisioned instances and excessive egress generate surprise bills (Bittok, 2022). **Mitigation:** FinOps habits from day one: budget alerts, resource tagging, rightsizing, and reserved pricing for stable workloads.
3. **Vendor lock-in and Skills gap:** Deeper managed-service adoption makes provider migration expensive (Eliaçık, 2022), and cloud requires a different operating mindset - infrastructure-as-code, monitoring-first, shared responsibility (McHaney, 2021). **Mitigation:** prioritize portability (containers, standard databases) and invest in targeted upskilling. Manvi and Shyam (2021) highlight that chain dependency in virtualized environments is a key risk teams must understand to operate effectively.

4. Recommended Cloud Models for ABC Enterprise

Cloud service models sit on a control-versus-responsibility spectrum (McHaney, 2021). IaaS gives ABC full flexibility over compute and configuration, but it also means managing operating systems and scaling policies. PaaS reduces that burden by abstracting infrastructure management, allowing the team to focus on building and improving the application itself. SaaS, while efficient, offers limited customization and is therefore less appropriate for a start-up that must differentiate its digital platform.

Deployment models also affect scalability and automation. Private cloud increases control but sacrifices elasticity and cost efficiency. Hybrid cloud can support cloud bursting and workload portability, yet introduces architectural complexity that exceeds ABC's current operational maturity (Manvi & Shyam, 2021). For ABC - a fast-growing, customer-facing

platform - public cloud remains the most strategically aligned option due to its scalability, global availability, and built-in automation capabilities (Mell & Grance, 2011; Nishimura, 2022).

Recommendation: A blended approach: IaaS for flexible compute layers, and PaaS for managed databases, load balancing, autoscaling, and serverless functions. This balance preserves flexibility while embedding the rapid elasticity and measured service characteristics (Mell & Grance, 2011) directly into the infrastructure layer, without rebuilding operational maturity from scratch (Accenture Technology, 2020).

5. Recommended Cost Model

Cloud providers offer three levers:

1. Pay-as-you-go (maximum flexibility, highest unit price),
2. Reserved/committed pricing (discounts for baseline commitments),
3. Spot/preemptible (deep discounts for interruption-tolerant workloads) (Bittok, 2022; Amazon Web Services, n.d.-b).

Recommendation: A hybrid cost model - reserved capacity for stable customer-facing tiers (web/app, databases), pay-as-you-go autoscaling for demand spikes, and spot instances for background jobs and analytics pipelines. TCO modelling should include not just cloud fees but also migration cost, security tooling, and engineering effort (Bittok, 2022). As Bittok (2022) notes, cloud adoption is rarely about the cheapest bill - it's about better ROI: less downtime, faster launches, and automation that avoids linear headcount growth.

6. Cloud Provider Recommendation: AWS

The realistic shortlist is AWS, Azure, or GCP. All three cover the technical baseline, but differentiate by ecosystem strength (Nishimura, 2022). AWS is the strongest fit for ABC: the scenario already references Route 53 (an AWS service), implying existing AWS-aligned architecture patterns. AWS pioneered affordable cloud infrastructure for start-ups - S3 and EC2 launched in 2006 - and offers the broadest managed-service catalogue for automation (Nishimura, 2022; Amazon Web Services, n.d.-a).

Two essential AWS elements for ABC's automation requirements:

1) Elastic Load Balancing + Auto Scaling: keeps the app responsive during demand surges by distributing traffic and scaling instances automatically, directly implementing rapid elasticity (Mell & Grance, 2011; Amazon Web Services, n.d.).

2) AWS Lambda: serverless compute for event-driven automation: order events, payment webhooks, image processing, notifications. No servers to manage; costs align with actual execution (Amazon Web Services, n.d.). Together, these turn operational workflows into policy-driven, event-driven systems - reusable, modular services consistent with SOA principles (McHaney, 2021).

7. Conclusion

Cloud computing is the right strategic move for ABC: measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011) directly translate into business automation outcomes - scaling without procurement delays, operations without linear headcount growth, and faster time-

to-market (Accenture Technology, 2020). The recommended stack is public cloud deployment, blended IaaS/PaaS service model, hybrid cost model, and AWS as provider. Managed well, ABC's infrastructure stops being a bottleneck and becomes a competitive advantage - scaling as far as product-market fit allows (Eliaçık, 2022). That approach restructures operational capability into programmable, automated systems aligned with business growth.

8. Appendices

8.1. Appendix A – Cloud vs Traditional IT Comparison Table

Dimension	Traditional IT	Cloud (ABC Case)
Cost model	CAPEX heavy	OPEX elasticity
Scaling	Hardware procurement	Auto Scaling
Deployment speed	Weeks/months	Minutes
Automation	Manual ops	Policy-driven

8.2. Appendix B – High-Level Architecture Diagram

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no

8.3. Appendix C – Cost Model Breakdown Example

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no

8.4. Appendix D – Shared Responsibility Model Diagram

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no

8.5. Appendix E – NIST Five Essential Characteristics Mapped to ABC

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no

8.6. Appendix F – Glossary of Terms

Term	Meaning
CAPEX	Capital Expenditure
OPEX	Operational Expenditure
IAM	Identity and Access Management
TCO	Total Cost of Ownership
SOA	Service-Oriented Architecture
PaaS	Platform as a Service
IaaS	Infrastructure as a Service
SaaS	Software as a Service

End of Appendix Section

9. References

Accenture Technology. (2020, June 5). *Why cloud matters* [Video]. YouTube. https://www.youtube.com/watch?v=p1Nr03gtkyU

Amazon Web Services. (n.d.-a). *AWS Well-Architected Framework.* https://aws.amazon.com/architecture/well-architected/

Amazon Web Services. (n.d.-b). *AWS Pricing.* https://aws.amazon.com/pricing/

Bittok, T. (2022, January 22). *Cloud total cost of ownership.* LinkedIn Pulse. https://www.linkedin.com/pulse/cloud-total-cost-ownership-theophilus-bittok/

Eliaçık, E. (2022). *The good, bad, and ugly sides of cloud computing.* Dataconomy. https://dataconomy.com/2022/05/pros-and-cons-of-cloud-computing-2022/

Manvi, S., & Shyam, G. K. (2021). *Cloud computing: Concepts and technologies* (Chapter 4). CRC Press. https://learning-oreilly-com.torrens.idm.oclc.org/library/view/cloud-computing/9781000338058/

McHaney, R. (2021). *Cloud technologies: An overview of cloud computing technologies for managers.* Wiley. https://ieeexplore-ieee-org.torrens.idm.oclc.org/servlet/opac?bknumber=9820907

Mell, P., & Grance, T. (2011). *The NIST definition of cloud computing* (Special Publication 800-145). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-145

Nishimura, H. (2022, August 30). *Introduction to AWS for non-engineers: 1 cloud concepts* [Video]. LinkedIn Learning. https://www.linkedin.com/learning/introduction-to-aws-for-non-engineers-1-cloud-concepts-2/

Rahman, M. S., & Raza, M. (2021). Cloud computing security challenges and its potential solution. *International Journal of Computer Applications, 174*(6), 29–33. https://doi.org/10.5120/ijca2021921167

Singh, S., & Chana, I. (2016). Cloud resource provisioning: Survey, status and future research directions. *Knowledge and Information Systems, 49*(3), 1005–1069. https://doi.org/10.1007/s10115-016-0922-3